# Week 3 | Module A Microphone Candidate Selection

Smart Digital Stethoscope | EEE 495 | Prepared by: Kaan
6 October 2026 | Version 2.0 | Technical selection for comparative testing

**This work is a student project and makes no diagnostic claim.** Physical testing will use an acoustic phantom only. No human recordings or physical test results are reported here. [1,3]

## 1) Weekly task and selection decision

The advisor's Week 3 Module A task is to study electret and MEMS microphones and select two candidates for comparison. I select **Kingstate KECG2740PBJ** and **PUI DMM-4026-B-I2S-R, supplied on the DMM-4026-B-I2S-EB-R evaluation board**, as the experimental pair. This selects the models to compare; it does not yet select the final system microphone or release either part for purchase. [1,4-6]

**Progress from Week 2:** the selection criteria and provisional models are now developed into a documented experimental pair, with a sensor/board specification check, an alternative screened for low-frequency suitability, and explicit acquisition-interface and Week 4 purchase conditions. No new measurements are claimed.

## 2) Why this pair

**Electret:** the Kingstate capsule provides an analogue path whose bias, gain and filtering can be characterized separately. Its pin package supports a removable acoustic holder. **MEMS:** the PUI board provides a digital I2S path and avoids soldering the bare miniature sensor. Its typical response plot extends to 20 Hz, making it worth evaluating for the project's low-frequency requirement. Neither advantage proves adequate phantom performance. [4-6]

Electret microphones use a charged condenser structure; MEMS refers to micromachined sensor construction and does not itself imply digital output. This PUI part is digital. The experiment therefore compares the **complete acquisition paths**, including electronics and mounting, rather than attributing every difference to the sensing element alone.

## 3) Datasheet evidence for the selected models

| Property | Kingstate KECG2740PBJ [4] | PUI sensor / EB-R board [5,6] |
|---|---|---|
| Output / supply | Analogue; standard supply 2 V; test load 2.2 kOhm. | I2S; rated 1.8 V, operating range 1.5-3.6 V. |
| Sensitivity | -40 +/-3 dB re 1 V/Pa, at 1 kHz and 1 Pa; nominal 10 mV/Pa. | Listed -26 +/-1 dB, but with a voltage-reference label despite digital output; clarification needed. |
| Catalogue SNR | 56 dBA at 1 kHz, 1 Pa. | 64 dBA at 1 kHz, 94 dB SPL. |
| Frequency evidence | Listed 20 Hz-20 kHz; plotted response begins near 50 Hz. | Listed 20 Hz-20 kHz; typical response plot extends to 20 Hz. |
| Lower -3 dB point | Not explicitly specified. | Not explicitly specified. |
| Physical form | Pin capsule, diameter 6 mm, height 2.7 mm. | EB-R board: 20 x 18 x 2 mm; preserve acoustic-port clearance. |

Catalogue SNR is A-weighted and does not establish the noise performance in our 20-500 Hz band. The 8 dB catalogue difference is not a predicted improvement in the completed device. [1,4,5]

# Low-Frequency Evidence and Integration

Week 3 | Selection rationale, limitations and Module A handoffs

## 4) Low-frequency requirement and unresolved evidence

The project characterizes 20-500 Hz; the component document particularly emphasizes 20-150 Hz. It requires recording each candidate's lower -3 dB point before ordering. A range label beginning at 20 Hz is insufficient to establish that point. The cutoff is defined relative to a stated passband reference: -3 dB corresponds to approximately 0.708 of the reference amplitude. [1,2]

**Kingstate warning:** the datasheet's -3 dB sensitivity-change entry concerns reducing supply voltage from 2 V to 1.5 V; it is not a frequency cutoff. The plotted curve does not establish the response from 20 to 50 Hz. **PUI:** the typical curve provides low-frequency evidence but no explicit guaranteed cutoff. Neither missing number has been replaced with an invented value. [4,5]

**Alternative screened:** TDK INMP441 specifies a flat response from 60 Hz to 15 kHz. It does not establish flat response in the important 20-50 Hz region, so I do not select it as a direct substitute solely because a seller may deliver it sooner. This does not mean it cannot detect signals below 60 Hz. [7]

## 5) Acquisition paths and interface conditions

| Path | Module A requirement and Module B handoff |
|---|---|
| Kingstate analogue | Provide capsule bias, low-noise AFE and an agreed ADC path. Record bias point, gain, ADC range, coupling capacitors and all high-pass corners. AFE/ADC and MCU choices remain open; analogue output cannot be wired directly to an I2S input. |
| PUI digital | Connect supply/ground and the documented I2S clock, word-select, data and channel-select signals. Ozan confirms pin mapping, levels, timing and sample alignment. The sensor specifies a 24-bit data field with 18-bit precision and a 32-bit word; these are different quantities. [5] |
| Clock proposal | Retain the REV01 proposal of 48 kHz with two 32-bit slots: BCLK = 48,000 x 2 x 32 = 3.072 MHz; retain one selected channel. The datasheet clock tables differ (page 1: 2.048-4.096 MHz; page 3 maximum: 12.288 MHz). This proposed clock fits both ranges, but timing still needs confirmation. [5,8] |

Sensitivity requires a reference and units. The PUI sensitivity label must not silently be treated as dBFS. For Kingstate, 10 mV/Pa at 1 kHz is a reference sensitivity, not a measured heart-sound voltage. Week 4 gain and headroom will be tied to the actual signal, ADC range and contact transients; a universal phantom amplitude has not been established.

## 6) Phantom and repeatable mounting

Prepare an exciter/small transducer, compatible amplifier, silicone or gel coupling layer, rigid frame and indexed chestpiece holder. Check transducer impedance, amplifier drive/clipping limits and useful output at 20-50 Hz. Preserve the REV01 parts as proposals until checked. Record layer thickness, compression, cavity geometry, contact pressure and cable restraint. [2,8]

Use the same phantom, excitation and controlled placement for both candidates. Record a quiet baseline and 20-500 Hz sweep; test electrical AFE response separately. Without an independent acoustic reference, report the combined phantom/acquisition response rather than claiming an isolated microphone response. Control motion/contact noise, ambient sound, 50 Hz pickup and changing mechanical coupling. Keep raw data before filtering.

# Week 4 Actions and Evidence Record

Week 3 | Coordination with Ozan and Ege | No purchase or team approval recorded

## 7) Conditions to close before ordering in Week 4

| Owner | Required action and evidence |
|---|---|
| Kaan | Obtain traceable low-frequency evidence for each exact part and record its lower -3 dB point and reference. If unavailable, keep the component instruction unresolved and take an alternative or a documented exception to the advisor. Resolve the PUI reference/timing ambiguities with Ozan. |
| Kaan + Ozan | Check AFE/ADC, board fit and connections. Reconcile the REV01 BOM: its ECM technical-reference entry and procurement entry must identify the selected model consistently. MCU/dev-board selection is not finalized in this report. |
| Ozan + team | Check laboratory stock first: dev-board, breadboard, exciter, function generator, oscilloscope and USB audio interface. Obtain an exact-part quote and confirmed arrival date. Plan two units per microphone model, including a spare, less suitable stock already available. [2] |
| Ege + Ozan | Ege owns implementation/validation of quality metrics on labeled recordings; Ozan owns the first device-PC frame draft. Agree sample rate, representation/scaling, path ID, gain/filter metadata, continuity counter and status information. Field sizes, byte order and transport remain B/C decisions. [1,8] |

**Sourcing routes for Ozan to confirm:** Kingstate through SAMM/Farnell; the exact PUI EB-R board through DigiKey, with its official Turkish reseller EKOM as a quotation/import route. A Turkish website does not establish domestic warehouse stock. No price, stock reservation or delivery promise is asserted here. [9]

## 8) Outcome and next milestone

The Week 3 Module A output is the selected experimental pair and its documented technical rationale. Purchase readiness remains open because the lower-cutoff evidence and integration checks are incomplete. Ordering is planned for **Week 4**, alongside the gain, bandwidth, noise-floor and clipping-headroom requirements. Ozan and Ege have not yet finalized joint decisions. [1,2]

The later evidence sequence remains: first end-to-end recorded signal by Week 5; phantom characterization and independently checked feasibility by Weeks 8-9; sensor comparison in Week 10 and at least ten repeat measurements in Week 11. The final microphone choice will follow those results. Save reviewed datasheets, decision changes and later raw data/scripts in the shared repository with genuine contribution history. [1,3]

## References

[1] Advisor, SDP4_Digital_Stethoscope_Plan_TR-1.docx: scope and EEE 495 weekly Module A/B/C rows. [2] SDP4_Component_Requirements_EN.docx: component quantities and Notes Before Ordering. [3] SDP_Assessment_Criteria_for_Students.docx: evidence, individual contribution and reporting rules.

[4] Kingstate, KECG2740PBJ, specifications and typical response, pp. 2-3. Manufacturer datasheet, Farnell-hosted.

[5] PUI Audio, DMM-4026-B-I2S-R, electrical specifications, I2S timing and response, pp. 1-3. Sensor datasheet.

[6] PUI Audio, DMM-4026-B-I2S-EB-R. Evaluation-board datasheet; official board specifications/dimensions.

[7] TDK InvenSense, INMP441, DS-INMP441-00, Rev. 1.1. Manufacturer datasheet, stated flat-response range.

[8] Team, Ozan_REV01_System_Design.docx and Ozan_REV01_BOM_Test.xlsx, 30 September 2026: architecture and BOM proposals. Team task lists KAAN-A1/A2 and TAKIM-01/02/03. Latest scheduling instruction: order in Week 4.

[9] Procurement routes: SAMM Kingstate; DigiKey PUI board; DigiKey official EKOM page. Reviewed 6 October 2026; item-specific quotation still required.